

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Wrote systems-level infrastructure code in Rust, Go, and C++ for scheduling and resource-allocation logic under GPU-scarcity constraints, working close to the metal on performance-sensitive paths.
- Built and maintained internal libraries powering agentic AI workflows (OpenAI Agents SDK, MCP tool-use) consumed by multiple teams, keeping interfaces stable while the underlying model providers and tool-calling formats changed.
- Worked directly with LLMs across providers (GPT, Claude, Llama, Gemini), reasoning about prompt structure, context formatting, and tool-call schemas to keep outputs reliable across model and API versions.
- Ran systematic experiments comparing retrieval, prompting, and tool-calling strategies, measuring output quality and latency before shipping changes to production.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Rebuilt a production RAG retrieval pipeline, running controlled experiments on chunking and representation choices that moved recall 61% to 89% on a 2K-query eval set.
- Optimized inference-serving performance (TensorRT-LLM) and cut p99 latency 420ms to 140ms across 50K+ daily requests through a systems-level async redesign.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Carried out a cross-cutting refactor standardizing failure-handling and API contracts across 25+ production Java/C++ microservices, reducing production escalations 60% without breaking downstream consumers.
- Architected and operated distributed systems for financial workflows serving 1M+ users at 99.9% uptime over 4 years on AWS.

PROJECTS

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Production LLM/agent evaluation platform measuring rubric-based output quality, drift, and regression across model and prompt-format versions; FastAPI backend, PostgreSQL.

LangFetch

tinyurl.com/LangFetch

- Agent decomposing natural-language queries into dependency-ordered SQL subqueries with MCP-compatible tool-use; designed the tool-call schema and context-passing abstraction used across subqueries.

Distributed KV Store

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus distributed storage engine in C++ handling concurrent writes with linearizable reads and log compaction, built for correctness under structural, low-level constraints.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Python, Rust, Go, C++, Java, SQL

LLM/Agentic Systems: OpenAI Agents SDK, MCP, LangGraph, Anthropic/Claude, RAG, Prompt/Context Engineering, Tool-Calling Schemas

ML: PyTorch, TensorFlow, Hugging Face, vLLM, TensorRT-LLM, Reinforcement Learning

Systems: Distributed Systems, API/Library Design, Large-Scale Refactoring, Performance Optimization, AWS, Docker

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer